MATH 130

Geometry: Circles & Solids

Circumference, Area, Arc Length, Sectors, Radians, Velocity, and Solid Figures

Liberty University | Spring 2026

Learning Goals

By the end of this lesson, you will be able to:

1. Identify circle components

Name and define radius, diameter, chord, tangent, secant, arc, sector, and segment.

2. Apply angle theorems in circles

Use inscribed angle, intersecting chord, and supplementary angle theorems to find unknown measures.

3. Convert angles (DMS & radians)

Convert between DMS, decimal degrees, and radians using π rad = 180° and DMS rules.

4. Calculate arc length and sector area

Apply $s = r\theta$ for arc length and $A = \frac{1}{2}r^2\theta$ for sector area (θ in radians).

5. Solve linear and angular velocity problems

Relate $v = r\omega$ and $\omega = \theta/t$; apply to real-world rotation scenarios.

6. Compute volume and surface area of solids

Use standard formulas for prisms, cylinders, cones, spheres, and pyramids.

SECTION 1

Circle Fundamentals

Definitions, terminology, and the key parts of every circle.

Circle Terminology

A circle is the set of all points in a plane that are equidistant from a fixed point called the center. The distance from the center to any point on the circle is the radius.

Radius

Distance from the center to any point on the circle. Half the diameter.

Diameter

A chord passing through the center. Equal to 2r (twice the radius).

Chord

A line segment with both endpoints on the circle.

Tangent

A line that touches the circle at exactly one point.

Secant

A line that intersects the circle at two points.

Center

The fixed point equidistant from all points on the circle.

Arcs, Sectors & Segments

Arc

A portion of the circle's circumference. A minor arc is less than a semicircle; a major arc is greater.

Central Angle

An angle formed at the center by two radii. Its measure equals the intercepted arc.

Sector

The region bounded by two radii and the arc they intercept — like a "slice of pie."

Segment

The region between a chord and its arc. A chord creates both a minor and a major segment.

Inscribed Angles

An inscribed angle has its vertex on the circle and its sides are chords. The key theorem relates it to its intercepted arc.

Definition

An inscribed angle of an arc is an angle whose vertex lies on the circle, whose sides pass through the endpoints of the arc, and whose vertex is not an endpoint of the arc.

Inscribed Angle Theorem

The measure of an inscribed angle is exactly one-half the measure of its intercepted arc.

$$\angle ABC = \tfrac{1}{2} \cdot \text{arc } AC$$

Corollary: An inscribed angle in a semicircle is always a right angle (90°), because the intercepted arc is 180°.

Chord & Angle Theorems

When two chords intersect inside a circle, the angles formed relate to the intercepted arcs. These theorems work together with supplementary angle rules.

Intersecting Chords Angle

$$\angle = \tfrac{1}{2}(\mathrm{arc}_1 + \mathrm{arc}_2)$$

When two chords cross inside a circle, the measure of the angle formed equals half the sum of the two intercepted arcs.

Vertical & Supplementary

Vertical angles are congruent

$\angle AEB = \angle DEC$ and $\angle AED = \angle BEC$

Adjacent angles are supplementary

$\angle AEB + \angle AED = 180°$

Inscribed Angle → Arc

If you know an inscribed angle, the intercepted arc = 2 × (angle). If you know the arc, the inscribed angle = ½ × (arc).

Diameter = 180° Arc

A diameter divides the circle into two semicircles of 180° each. If AC is a diameter, then arc ABC = 180° and arc ADC = 180°.

Worked Example: Circle Angles

∠AED = 133° and arc AB = 52°. AC is a diameter (BD is not). Find ∠AEB, ∠ACB, and arc CD.

Part (a): Find ∠AEB

∠AEB and ∠AED are supplementary (they form a straight line).

∠AEB = 180° − 133° = 47°

Part (b): Find ∠ACB

∠ACB is an inscribed angle that intercepts arc AB = 52°.

∠ACB = ½ × 52° = 26°

Part (c): Find arc CD

∠AED = 133° is formed by intersecting chords, so:

133° = ½(arc AD + arc BC). Since AC is a diameter:

arc ABC = 180° → arc BC = 180° − 52° = 128°

So: 133° = ½(arc AD + 128°) → 266° = arc AD + 128°

arc AD = 138°

arc ADC = 180° (semicircle) → arc CD = 180° − 138° = 42°

(a) ∠AEB

= 47°

(b) ∠ACB

= 26°

(c) arc CD

= 42°

Circumference & Area

Circumference

$$C = \pi d = 2\pi r$$

The distance around a circle. Use $\pi \approx 3.14159$ for decimal answers, or leave in terms of π for exact answers.

Area

$$A = \pi r^2$$

The area enclosed by the circle. Always use the radius (not the diameter) in this formula.

Example: Circumference

Find the circumference for $r = 1$, $r = 5$, $r = \pi/2$, and $r = 7.6$.
Solutions: $C = 2\pi$, $C = 10\pi$, $C = \pi^2$, $C \approx 47.752$

SECTION 2

Angle Measure & Arc Length

DMS notation, radians, degree–radian conversion, arc length, and sector area formulas.

Degrees, Minutes & Seconds

Angles in navigation and surveying often use DMS (degrees-minutes-seconds) notation. 1° = 60′ (minutes), 1′ = 60″ (seconds).

DMS → Decimal Degrees

$$\text{Decimal} = D + M/60 + S/3600$$

Divide minutes by 60 and seconds by 3600, then add all three parts together.

Decimal → DMS

1. Whole part = degrees
2. Decimal × 60 → whole part = minutes
3. Remaining decimal × 60 = seconds

Example: 37°41′11″ → Decimal

$= 37 + 41/60 + 11/3600$
$= 37 + 0.6833 + 0.003056$
$= 37.6864°$

Example: 36.3492° → DMS

D = 36°. $0.3492 \times 60 = 20.952 \rightarrow M = 20'$
$0.952 \times 60 = 57.12 \rightarrow S \approx 57''$
Result: 36°20′57″

Degrees & Radians

If a central angle intercepts an arc equal in length to the radius, that angle measures exactly 1 radian. The radius can be marked off along the circumference approximately $2\pi \approx 6.28$ times.

Key Relationship

$$2\pi \text{ rad} = 360^\circ$$
$$\pi \text{ rad} = 180^\circ$$

Degrees → Radians

Multiply by π
Divide by 180

Radians → Degrees

Multiply by 180
Divide by π

One radian ≈ 57.296°. Think of it as the angle that "unwraps" one radius-length of arc.

Conversion Examples

Radians → Degrees

$\pi/6 \rightarrow (\pi/6)(180/\pi) = 30°$

2.7 rad $\rightarrow (2.7)(180/\pi) \approx 154.699°$

1 rad $\rightarrow (1)(180/\pi) \approx 57.296°$

Degrees → Radians

$45° \rightarrow (45)(\pi/180) = \pi/4$

$120° \rightarrow (120)(\pi/180) = 2\pi/3$

$255° \rightarrow (255)(\pi/180) = 17\pi/12$

$1° \rightarrow (1)(\pi/180) = \pi/180 \approx 0.01745$ rad

Remember: π rad = 180° is the bridge. Multiply by whichever fraction cancels the unit you want to eliminate.

Mil: A unit used in artillery — 1 mil = 1/6400 of a full circle. To convert degrees to mils: multiply by $6400/360 \approx 17.78$.

Arc Length

Arc length is the distance along the curved part of a circle between two points. The central angle must be in radians.

$$s = r\theta$$

s = arc length, r = radius, θ = central angle (in radians)

If θ is in degrees:

Convert to radians first by multiplying by $\pi/180$, then apply $s = r\theta$.

Example 1

Central angle = $\pi/3$, radius = 10.
$s = r\theta = 10 \cdot (\pi/3) = 10\pi/3 \approx 10.472$

Example 2

Central angle = 45°, radius = 21.
Convert: $45° \times \pi/180 = \pi/4$
$s = 21 \cdot (\pi/4) = 21\pi/4 \approx 16.493$

Area of a Sector

A sector is the "pie-slice" region between two radii and an arc. Its area is proportional to its central angle.

$$A = \tfrac{1}{2} r^2\theta$$

A = area, r = radius, θ = central angle (in radians)

Alternatively (degrees):

$A = (\theta/360) \cdot \pi r^2$ where θ is in degrees. This is simply the fraction of the full circle's area.

Example 1

Central angle = $\pi/3$, radius = 10.
$A = \tfrac{1}{2}(10^2)(\pi/3) = 50\pi/3 \approx 52.360$

Example 2

Central angle = $45°$, radius = 21.
Convert: $45° \times \pi/180 = \pi/4$
$A = \tfrac{1}{2}(21^2)(\pi/4) = 441\pi/8 \approx 173.18$

Applied Example: Aviation Arc

You are instructed to arc south on the 12 NM arc from the 090 radial to the 210 radial.

Part A — Arc Distance

Central angle: 210° − 90° = 120° → convert to radians: $120 \times \pi/180 = 2\pi/3$.
Arc length: $s = r\theta = 12 \cdot (2\pi/3) = 8\pi \approx 25.13$ NM.

Part B — Time at 120 Knots

Time = Distance / Speed = 25.13 NM / 120 kt ≈ 0.209 hr ≈ 12.6 minutes.

"That Looks About Right" — pilots use quick mental math to verify computed results. 120° is one-third of a full circle, so the arc should be about one-third of the circumference: $\frac{1}{3} \times 2\pi(12) \approx 25$ NM. ✓

SECTION 3

Linear & Angular Velocity

Connecting rotational speed to the speed of a point on a spinning object.

Linear & Angular Velocity

Angular Velocity (ω)

$$\omega = \theta / t$$

The rate of rotation — how fast the angle changes. Measured in radians per unit time (rad/s, rad/min).

Linear Velocity (v)

$$v = r\omega = s / t$$

The speed of a point on the rotating object. Farther from the center → faster linear speed for the same ω.

Converting RPM to Radians

1 revolution = 2π radians. So n RPM = $n \times 2\pi$ rad/min. Always convert revolutions to radians before using $v = r\omega$.

Key insight: $v = r\omega$ shows that linear velocity is directly proportional to both radius and angular velocity.

$s = r\vartheta$ → $v = s/t = r(\vartheta/t) = r\omega$ → $v = r\omega$ *These three formulas are all connected.*

Worked Example: Riverboat Paddles

The paddles of a riverboat have a radius of 8.50 ft and turn at 20.0 revolutions per minute (RPM). What is the speed of a tip of one paddle?

Solution

Step 1 — Convert RPM to angular velocity:
$\omega = 20.0$ rev/min × 2π rad/rev = 40π rad/min

Step 2 — Apply $v = r\omega$:
$v = 8.50$ ft × 40π rad/min = 340π ft/min

Step 3 — Convert to convenient units:
$v = 340\pi \approx 1068$ ft/min ≈ 12.14 mi/hr

≈ 1068 ft/min

≈ 12.14 mi/hr

The paddle tip moves at about 12 miles per hour through the water.

More Velocity Examples

Drive Shaft Rotation

A car's tachometer reads 3200 RPM. Through what total angle does the drive shaft rotate in 1 second?

$\omega = 3200 \text{ rev/min} \times (2\pi \text{ rad/rev}) \times (1 \text{ min}/60 \text{ s})$
$\quad = 3200 \times 2\pi/60 \approx 335.1 \text{ rad/s}$
In 1 s: $\theta = 335.1 \text{ rad} \approx 53.3$ revolutions

Wind Turbine Blade Tips

GE 1.5 MW turbine: blades are 116 ft long, spinning at 22.0 RPM. Find the tip speed.

$\omega = 22.0 \times 2\pi = 44\pi \text{ rad/min}$
$v = r\omega = 116 \times 44\pi \approx 16{,}035 \text{ ft/min}$
≈ 182.2 mph

Bonus: Turbine Sweep Area

The area swept by the rotating blades forms a circle:
$A = \pi r^2 = \pi(116)^2 = 13{,}456\pi \approx 42{,}273 \text{ ft}^2$

Convert to acres: $42{,}273 \text{ ft}^2 \div 43{,}560 \text{ ft}^2/\text{acre} \approx 0.97$ acres — nearly a full acre of wind capture.

SECTION 4

Solid Geometric Figures

Volume and surface area formulas for common three-dimensional shapes.

Volume & Surface Area

Rectangular Prism

Volume

$V = lwh$

Surface Area

$SA = 2(lw + lh + wh)$

Cylinder

Volume

$V = \pi r^2 h$

Surface Area

$SA = 2\pi r^2 + 2\pi rh$

Cone

Volume

$V = \frac{1}{3}\pi r^2 h$

Surface Area

$SA = \pi r^2 + \pi rl$

Sphere

Volume

$V = \frac{4}{3}\pi r^3$

Surface Area

$SA = 4\pi r^2$

Right Pyramid

Volume

$V = \frac{1}{3}Bh$

Surface Area

$SA = B + \frac{1}{2}Pl$

Triangular Prism

Volume

$V = Bh$

Surface Area

$SA = 2B + Ph$

B = base area, P = base perimeter, l = slant height, h = height, r = radius.

Worked Example: Drain Pipe

A 12 ft long drain pipe (inner diameter = 4 inches) is clogged. Will two 5-gallon buckets hold the water pumped out? (1 gallon = 231 in^3)

Solution

Step 1 — Convert to consistent units:
Length: 12 ft = 144 in. Radius: diameter/2 = 4/2 = 2 in.

Step 2 — Volume of the cylindrical pipe:
$V = \pi r^2 h = \pi(2^2)(144) = 576\pi \approx 1809.56 \text{ in}^3$

Step 3 — Convert to gallons:
$V = 1809.56 \div 231 \approx 7.83$ gallons

Step 4 — Compare:
Two 5-gallon buckets = 10 gallons. Since $7.83 < 10$, yes — two buckets are sufficient.

Yes ✓

≈ 7.83 gal

Two 5-gallon buckets provide more than enough capacity (10 gal > 7.83 gal).

References

Katz, V. J. (1998). A History of Mathematics: An Introduction (2nd ed.). New York, NY: Addison-Wesley.